

Task-1: Find the root of the equation $x^3-x-2=0$ between [1,2] using Bisection Method:

Code (Python):

```
# for the equation only f(x)=x^3-x-2
from prettytable import PrettyTable

Co_efficient=[-2,-1,0,1]

def func_value(val):
    res=Co_efficient[0]
    for i in range(1,4):
        res+=(val**i)*Co_efficient[i]
    return res

table=PrettyTable(["Iter","a","b","Xr","f(a)","f(b)","f(Xr)"])
table.float_format=".4"
i=0
a=1
b=2
e=0.0001
p_Xr=0
for i in range(0,20):
    aa=func_value(a)
    bb=func_value(b)
    Xr=(a+b)/2
    F_Xr=func_value(Xr)
    table.add_row([i+1,a,b,Xr,aa,bb,F_Xr])
    if F_Xr==0:
        break
    if aa*F_Xr<0:
        b=Xr
    else:
        a=Xr
    if abs(abs(p_Xr)-abs(Xr))<=e:
        break
    p_Xr=Xr
print(table)
print(f"The root is {Xr:.4f} between that interval\n")
```

Output:

Iter	a	b	Xr	f(a)	f(b)	f(Xr)
1	1	2	1.5000	-2	4	-0.1250
2	1.5000	2	1.7500	-0.1250	4	1.6094
3	1.5000	1.7500	1.6250	-0.1250	1.6094	0.6660
4	1.5000	1.6250	1.5625	-0.1250	0.6660	0.2522
5	1.5000	1.5625	1.5312	-0.1250	0.2522	0.0591
6	1.5000	1.5312	1.5156	-0.1250	0.0591	-0.0341
7	1.5156	1.5312	1.5234	-0.0341	0.0591	0.0123
8	1.5156	1.5234	1.5195	-0.0341	0.0123	-0.0110
9	1.5195	1.5234	1.5215	-0.0110	0.0123	0.0006
10	1.5195	1.5215	1.5205	-0.0110	0.0006	-0.0052
11	1.5205	1.5215	1.5210	-0.0052	0.0006	-0.0023
12	1.5210	1.5215	1.5212	-0.0023	0.0006	-0.0008
13	1.5212	1.5215	1.5214	-0.0008	0.0006	-0.0001
14	1.5214	1.5215	1.5214	-0.0001	0.0006	0.0003

The root is 1.5214 between that interval

Task-2: Find the root of any Polynomial equation using Bisection Method:

Code (Python):

```
# for the general polynomial equation
from prettytable import PrettyTable
Co_efficient=[]

table=PrettyTable(["Iter","a","b","Xr","f(a)","f(b)","f(Xr)"])
table.float_format=".5"

def func_value(val):
    res=Co_efficient[0]
    for i in range(1,len(Co_efficient)):
        res+=(val**i)*Co_efficient[i]
    return res

def check(x,y):
    return (y<0 and x>0) or (x<0 and y>0)

def Bisection_Func(a,b):
    i=1
    p_Xr=0
    while True:
        aa=func_value(a)
        bb=func_value(b)
        Xr=(a+b)/2
        F_Xr=func_value(Xr)
        table.add_row([i,a,b,Xr,aa,bb,F_Xr])
```

```

        if F_Xr==0:
            return Xr
        if aa*F_Xr<0:
            b=Xr
        else:
            a=Xr
        if abs(abs(p_Xr)-abs(Xr))<=e:
            return Xr
        p_Xr=Xr
        i+=1

N=int(input("What is the Highest Degree of the Equation: "))
for i in range(N,0,-1):
    co_eff=int(input(f"The Co-efficient of X^{i} is: "))
    Co_efficient.append(co_eff)
Co_efficient.append(int(input("The Constant Value: \n")))
Co_efficient=Co_efficient[::-1]

e=0.0001
a=0
b=0
flag=0
F_a=func_value(a)
for i in range(1,51):
    F_b=func_value(i)
    if check(F_a,F_b):
        b=i
        flag=1
        break
    F_b=func_value(-i)
    if check(F_a,F_b):
        b=-i
        flag=1
        break
    i+=1

if flag:
    Xr=Bisection_Func(a,b)

    print(table)
    print(f"A root {Xr:.5f} is found between [{a},{b}] interval\n")
else:
    print("No such interval found")

```

Output:

What is the Highest Degree of the Equation: 3

The Co-efficient of X^3 is: 1

The Co-efficient of X^2 is: 0

The Co-efficient of X^1 is: -1

The Constant Value: -2

Iter	a	b	Xr	f(a)	f(b)	f(Xr)
1	0	2	1.00000	-2	4	-2.00000
2	1.00000	2	1.50000	-2.00000	4	-0.12500
3	1.50000	2	1.75000	-0.12500	4	1.60938
4	1.50000	1.75000	1.62500	-0.12500	1.60938	0.66602
5	1.50000	1.62500	1.56250	-0.12500	0.66602	0.25220
6	1.50000	1.56250	1.53125	-0.12500	0.25220	0.05911
7	1.50000	1.53125	1.51562	-0.12500	0.05911	-0.03405
8	1.51562	1.53125	1.52344	-0.03405	0.05911	0.01225
9	1.51562	1.52344	1.51953	-0.03405	0.01225	-0.01097
10	1.51953	1.52344	1.52148	-0.01097	0.01225	0.00062
11	1.51953	1.52148	1.52051	-0.01097	0.00062	-0.00518
12	1.52051	1.52148	1.52100	-0.00518	0.00062	-0.00228
13	1.52100	1.52148	1.52124	-0.00228	0.00062	-0.00083
14	1.52124	1.52148	1.52136	-0.00083	0.00062	-0.00010
15	1.52136	1.52148	1.52142	-0.00010	0.00062	0.00026

A root 1.52142 is found between [0,2] interval

Task-3: Find the root of the equation $xe^x - 1 = 0$ between [0,1] using Bisection Method:

Code (Python):

```
# for the equation only f(x)=xe^x - 1
from prettytable import PrettyTable
import math

def func_value(val):
    res=val*(math.e**val)-1
    return res

table=PrettyTable(["Iter","a","b","Xr","f(a)","f(b)","f(Xr)"])
table.float_format=".4"
i=0
a=0
b=1
e=0.0001
p_Xr=0
for i in range(0,20):
    aa=func_value(a)
    bb=func_value(b)
    Xr=(a+b)/2
    F_Xr=func_value(Xr)
```

```

table.add_row([i+1,a,b,Xr,aa,bb,F_Xr])
if aa*f_Xr<0:
    b=Xr
else:
    a=Xr
if abs(abs(p_Xr)-abs(Xr))<=e:
    break
p_Xr=Xr
print(table)
print(f"The root is {Xr:.4f} between that interval\n")

```

Output:

Iter	a	b	Xr	f(a)	f(b)	f(Xr)
1	0	1	0.5000	-1.0000	1.7183	-0.1756
2	0.5000	1	0.7500	-0.1756	1.7183	0.5878
3	0.5000	0.7500	0.6250	-0.1756	0.5878	0.1677
4	0.5000	0.6250	0.5625	-0.1756	0.1677	-0.0128
5	0.5625	0.6250	0.5938	-0.0128	0.1677	0.0751
6	0.5625	0.5938	0.5781	-0.0128	0.0751	0.0306
7	0.5625	0.5781	0.5703	-0.0128	0.0306	0.0088
8	0.5625	0.5703	0.5664	-0.0128	0.0088	-0.0020
9	0.5664	0.5703	0.5684	-0.0020	0.0088	0.0034
10	0.5664	0.5684	0.5674	-0.0020	0.0034	0.0007
11	0.5664	0.5674	0.5669	-0.0020	0.0007	-0.0007
12	0.5669	0.5674	0.5671	-0.0007	0.0007	-0.0000
13	0.5671	0.5674	0.5673	-0.0000	0.0007	0.0003
14	0.5671	0.5673	0.5672	-0.0000	0.0003	0.0002

The root is 0.5672 between that interval

Discussion:

If $f(x)$ is continuous between a , b point and if $f(a).f(b)<0$, then there lies at least one root between a , b . It is a very important method to find out the root of the polynomial or transcendental equation if the curve cut the x axis.

Rules of this method:

1. Find a & b such that $f(a).f(b)<0$
2. $X_r=(a+b)/2$
3. If $f(a).f(X_r)<0$ then $b= X_r$ go to step 2.
4. If $f(b).f(X_r)<0$ then $a= X_r$ go to step 2.
5. If $f(X_r)=0$, X_r is the root of the equation.